OS Orange Problem Report

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Class: 4-J

Phase 1

1A)

```
saketh@LAPTOP-MF8KSHCD:/mnt/c/Users/V Sai Sri Saketh/Desktop/OS Orange/PES2UG24CS602-pes-vcs$ make test_objects
./test_objects

make: 'test_objects' is up to date.
Stored blob with hash: d58213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
Object stored at: .pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
PASS: blob storage
PASS: deduplication
PASS: integrity check

All Phase 1 tests passed.
saketh@LAPTOP-MF8KSHCD:/mnt/c/Users/V Sai Sri Saketh/Desktop/OS Orange/PES2UG24CS602-pes-vcs$
```

1B)

```
saketh@LAPTOP-MF8KSHCD:/mnt/c/Users/V Sai Sri Saketh/Desktop/OS Orange/PES2UG24CS602-pes-vcs$ find .pes/objects -type f
.pes/objects/25/ef1fa07ea68a52f800dc80756ee6b7ae34b337afedb9b46a1af8e11ec4f476
.pes/objects/2a/594d39232787fba8eb7287418aec99c8fc2ecdaf5aaf2e650eda471e566fcf
.pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
saketh@LAPTOP-MF8KSHCD:/mnt/c/Users/V Sai Sri Saketh/Desktop/OS Orange/PES2UG24CS602-pes-vcs$
```

Phase 2

2A)

```
saketh@LAPTOP-MF8KSHCD:/mnt/c/Users/V Sai Sri Saketh/Desktop/OS Orange/PES2UG24CS602-pes-vcs$ make test_tree
./test_tree
gcc -Wall -Wextra -O2 -c test_tree.c -o test_tree.o
gcc -Wall -Wextra -O2 -c tree.c -o tree.o
gcc -o test_tree test_tree.o object.o tree.o -lcrypto
Serialized tree: 139 bytes
PASS: tree serialize/parse roundtrip
PASS: tree deterministic serialization

All Phase 2 tests passed.
saketh@LAPTOP-MF8KSHCD:/mnt/c/Users/V Sai Sri Saketh/Desktop/OS Orange/PES2UG24CS602-pes-vcs$
```

2B)

```
saketh@LAPTOP-MF8KSHCD:/mnt/c/Users/V Sai Sri Saketh/Desktop/OS Orange/PES2UG24CS602-pes-vcs$ xxd .pes/objects/48/9c3eba17e9892e8f0806f450161cba9442ea8b61a3
177305e6b63b3dabe272 | head -20
00000000: 636f 6d6d 6974 2032 3539 0074 7265 6520  commit 259.tree
00000010: 6464 3832 3334 3062 3565 3430 3365 3264  dd82340b5e403e2d
00000020: 6331 3862 6663 6130 3530 6536 3639 3939  c18bfca050e66999
00000030: 3335 3834 3937 3130 6232 6235 6432 3861  35849710b2b5d28a
00000040: 6238 6262 6666 6262 3732 6433 3236 3237  b8bbffbb72d32627
00000050: 0a70 6172 656e 7420 6165 6431 3038 3662  .parent aed1086b
00000060: 3935 6532 6336 6666 3032 3733 3138 6137  95e2c6ff027318a7
00000070: 3664 3262 6139 3239 6437 6264 6238 3162  6d2ba929d7bdb81b
00000080: 3964 3934 6132 6461 3566 6435 3466 3039  9d94a2da5fd54f09
00000090: 6230 3065 3263 3137 0a61 7574 686f 7220  b00e2c17.author
000000a0: 5620 5361 6920 5372 6920 5361 6b65 7468  V Sai Sri Saketh
000000b0: 203c 5045 5332 5547 3234 4353 3630 323e   <PES2UG24CS602>
000000c0: 2031 3737 3634 3439 3336 390a 636f 6d6d   1776449369.comm
000000d0: 6974 7465 7220 5620 5361 6920 5372 6920  itter V Sai Sri
000000e0: 5361 6b65 7468 203c 5045 5332 5547 3234  Saketh <PES2UG24
000000f0: 4353 3630 323e 2031 3737 3634 3439 3336  CS602> 177644936
00000100: 390a 0a41 6464 6564 2066 696c 6532       9..Added file2
saketh@LAPTOP-MF8KSHCD:/mnt/c/Users/V Sai Sri Saketh/Desktop/OS Orange/PES2UG24CS602-pes-vcs$
```

Phase 3

3A)

```
saketh@LAPTOP-MF8KSHCD:/mnt/c/Users/V Sai Sri Saketh/Desktop/OS Orange/PES2UG24CS602-pes-vcs$ make clean
make pes
./pes init
echo "hello" > file1.txt
echo "world" > file2.txt
./pes add file1.txt file2.txt
rm -f pes test_objects test_tree object.o tree.o index.o commit.o pes.o test_objects.o test_tree.o
rm -rf .pes
gcc -Wall -Wextra -O2 -c object.c -o object.o
gcc -Wall -Wextra -O2 -c tree.c -o tree.o
gcc -Wall -Wextra -O2 -c index.c -o index.o
gcc -Wall -Wextra -O2 -c commit.c -o commit.o
gcc -Wall -Wextra -O2 -c pes.c -o pes.o
gcc -o pes object.o tree.o index.o commit.o pes.o -lcrypto
Initialized empty PES repository in .pes/
```

3B)

```
saketh@LAPTOP-MF8KSHCD:/mnt/c/Users/V Sai Sri Saketh/Desktop/OS Orange/PES2UG24CS602-pes-vcs$ ./pes status
cat .pes/index
Staged changes:
  staged:     file1.txt
  staged:     file2.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  untracked:  .gitignore
  untracked:  commit.c
  untracked:  commit.h
  untracked:  index.c
  untracked:  index.h
  untracked:  Makefile
  untracked:  object.c
  untracked:  pes.c
  untracked:  pes.h
  untracked:  PHASE1_NOTES.md
  untracked:  README.md
  untracked:  test_objects.c
  untracked:  test_sequence.sh
  untracked:  test_tree.c
  untracked:  tree.c
  untracked:  tree.h

100755 2cf8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4 1776449067 6 file1.txt
100755 e00c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f 1776449067 6 file2.txt
saketh@LAPTOP-MF8KSHCD:/mnt/c/Users/V Sai Sri Saketh/Desktop/OS Orange/PES2UG24CS602-pes-vcs$
```

Phase 4

4A)

```
saketh@LAPTOP-MF8KSHCD:/mnt/c/Users/V Sai Sri Saketh/Desktop/OS Orange/PES2UG24CS602-pes-vcs$ make clean
make pes
./pes init

echo "Hello" > hello.txt
./pes add hello.txt
./pes commit -m "file1 added"

echo "World" >> hello.txt
./pes add hello.txt
./pes commit -m "Added file2"

echo "Goodbye" > bye.txt
./pes add bye.txt
./pes commit -m "third file"

./pes log
rm -f pes test_objects test_tree object.o tree.o index.o commit.o pes.o test_objects.o test_tree.o
rm -rf .pes
gcc -Wall -Wextra -O2 -c object.c -o object.o
gcc -Wall -Wextra -O2 -c tree.c -o tree.o
gcc -Wall -Wextra -O2 -c index.c -o index.o
gcc -Wall -Wextra -O2 -c commit.c -o commit.o
gcc -Wall -Wextra -O2 -c pes.c -o pes.o
gcc -o pes object.o tree.o index.o commit.o pes.o -lcrypto
Initialized empty PES repository in .pes/
Committed: aed1086b95e2... file1 added
Committed: 489c3eba17e9... Added file2
Committed: d8cbb50c0298... third file
commit d8cbb50c02987f2bc6a8961b0228fc4efe3627247b48d98f78e74d01d235c6f8
Author: V Sai Sri Saketh <PES2UG24CS602>
Date:   1776449369

    third filer auth!

commit 489c3eba17e9892e8f0806f450161cba9442ea8b61a3177305e6b63b3dabe272
Author: V Sai Sri Saketh <PES2UG24CS602>
Date:   1776449369

    Added file2 auth!
```

```
commit 489c3eba17e9892e8f0806f450161cba9442ea8b61a3177305e6b63b3dabe272
Author: V Sai Sri Saketh <PES2UG24CS602>
Date:   1776449369

    Added file2 auth!

commit aed1086b95e2c6ff027318a76d2ba929d7bdb81b9d94a2da5fd54f09b00e2c17
Author: V Sai Sri Saketh <PES2UG24CS602>
Date:   1776449369

    file1 added

saketh@LAPTOP-MF8KSHCD:/mnt/c/Users/V Sai Sri Saketh/Desktop/OS Orange/PES2UG24CS602-pes-vcs$
```

4B)

```
saketh@LAPTOP-MF8KSHCD:/mnt/c/Users/V Sai Sri Saketh/Desktop/OS Orange/PES2UG24CS602-pes-vcs$ find .pes -type f | sort
.pes/HEAD
.pes/index
.pes/objects/05/c8d777d0be85748b6788d1b5caf7d69754e6e68060c265ca21cbd6c05e5f43
.pes/objects/48/9c3eba17e9892e8f0806f450161cba9442ea8b61a3177305e6b63b3dabe272
.pes/objects/55/8c1061e21bccbeb23c527d8dcef5c8646cabfb6091aae3ae7ee3320c9ff7b7
.pes/objects/66/224663d23e6f4d9de9e2c7e6d8764305a92a3830a1a52d3d5f4aa8007b5c39
.pes/objects/ae/d1086b95e2c6ff027318a76d2ba929d7bdb81b9d94a2da5fd54f09b00e2c17
.pes/objects/cc/e49eb2943f54e8135f291862bdf218064409ba05a7ee7d5bec00ddc22addfb
.pes/objects/d8/cbb50c02987f2bc6a8961b0228fc4efe3627247b48d98f78e74d01d235c6f8
.pes/objects/dd/82340b5e403e2dc18bfca050e6699935849710b2b5d28ab8bbffbb72d32627
.pes/objects/e6/7ed66bcb4a708250a89f315d4d8bb92703343c84df834438880e13a68652bc
.pes/refs/heads/main
saketh@LAPTOP-MF8KSHCD:/mnt/c/Users/V Sai Sri Saketh/Desktop/OS Orange/PES2UG24CS602-pes-vcs$
```

4C)

```
saketh@LAPTOP-MF8KSHCD:/mnt/c/Users/V Sai Sri Saketh/Desktop/OS Orange/PES2UG24CS602-pes-vcs$ cat .pes/refs/heads/main
cat .pes/HEAD
d8cbb50c02987f2bc6a8961b0228fc4efe3627247b48d98f78e74d01d235c6f8
ref: refs/heads/main
```

FINAL INTEGRATION

```
saketh@LAPTOP-MF8KSHCD:/mnt/c/Users/V Sai Sri Saketh/Desktop/OS Orange/PES2UG24CS602-pes-vcs$ make test-integration
=== Running integration tests ===
bash test_sequence.sh
=== PES-VCS Integration Test ===

--- Repository Initialization ---
Initialized empty PES repository in .pes/
PASS: .pes/objects exists
PASS: .pes/refs/heads exists
PASS: .pes/HEAD exists

--- Staging Files ---
Status after add:
Staged changes:
  staged:     file.txt
  staged:     hello.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  (nothing to show)


--- First Commit ---
Committed: 229c35e8c2c0... Initial commit

Log after first commit:
commit 229c35e8c2c06e2643961d46d87e46bfdbf9bbe6468695576e7e7d565463a1fa
Author: V Sai Sri Saketh <PES2UG24CS602>
Date:   1776450132

    Initial commit


--- Second Commit ---
Committed: 4750d5fa7d2d... Update file.txt

--- Third Commit ---
Committed: 380cbc534c17... Add farewell
```

saketh@LAPTOP-MF8KSHCD:

```
--- Full History ---
commit 380cbc534c173c150fe5ab5a385b891e398d5bb38e513a6cc2ce5e4d1b156df8
Author: V Sai Sri Saketh <PES2UG24CS602>
Date:   1776450132

    Add farewellauth!

commit 4750d5fa7d2d35bcc567020ca9f1d8f062b7111b5d6c95bd55ce948b76f3b8de
Author: V Sai Sri Saketh <PES2UG24CS602>
Date:   1776450132

    Update file.txth!

commit 229c35e8c2c06e2643961d46d87e46bfdbf9bbe6468695576e7e7d565463a1fa
Author: V Sai Sri Saketh <PES2UG24CS602>
Date:   1776450132

    Initial commit


--- Reference Chain ---
HEAD:
ref: refs/heads/main
refs/heads/main:
380cbc534c173c150fe5ab5a385b891e398d5bb38e513a6cc2ce5e4d1b156df8

--- Object Store ---
Objects created:
10
.pes/objects/0b/d69098bd9b9cc5934a610ab65da429b525361147faa7b5b922919e9a23143d
.pes/objects/10/1f28a12274233d119fd3c8d7c7216054ddb5605f3bae21c6fb6ee3c4c7cbfa
.pes/objects/22/9c35e8c2c06e2643961d46d87e46bfdbf9bbe6468695576e7e7d565463a1fa
.pes/objects/38/0cbc534c173c150fe5ab5a385b891e398d5bb38e513a6cc2ce5e4d1b156df8
.pes/objects/47/50d5fa7d2d35bcc567020ca9f1d8f062b7111b5d6c95bd55ce948b76f3b8de
.pes/objects/58/a67ed1c161a4e89a110968310fe31e39920ef68d4c7c7e0d6695797533f50d
.pes/objects/ab/5824a9ec1ef505b5480b3e37cd50d9c80be55012cabe0ca572dbf959788299
.pes/objects/b1/7b838c5951aa88c09635c5895ef7e08f7fa1974d901ce282f30e08de0ccd92
.pes/objects/d0/7733c25d2d137b7574be8c5542b562bf48bafeaa3829f61f75b8d10d5350f9
.pes/objects/db/07a1451ca9544dbf66d769b505377d765efae7adc6b97b75cc9d2b3b3da6ff

=== All integration tests completed ===
```

Phase 5

Q 5.1

The command `pes checkout <branch>` basically switches the project to another branch. To do this, it first reads the file `.pes/refs/heads/<branch>` to get the commit hash of that branch. Then it updates `.pes/HEAD` so that it points to that branch.

After that, it loads the commit and its tree, and updates the working directory so that all files match the selected branch. This means creating new files, deleting unnecessary ones, and updating existing ones. It also updates the `.pes/index` to match the checked-out state.

This process is a bit tricky because the program must be careful not to overwrite any uncommitted changes. It has to compare the current files with the target branch and only make safe changes.

Q 5.2

To detect conflicts during checkout, we need to compare three things: the working directory, the index, and the target branch.

First, we check if a file in the working directory has been modified compared to the index. This can be done using metadata like size and time, or by recomputing the file hash if needed. Then we compare the index version of the file with the version in the target branch.

If the file has been modified locally and the target branch also has a different version of that same file, then there is a conflict. In that case, checkout should stop to avoid overwriting the user's changes.

Q 5.3

In a detached HEAD state, `.pes/HEAD` directly stores a commit hash instead of pointing to a branch.

If a user makes a commit in this state, the commit is created normally, but it is not attached to any branch. So no branch gets updated, and the commit can become difficult to find later.

However, it is still possible to recover it as long as the commit hash is known. The user can create a new branch pointing to that commit. This works as long as the commit has not been deleted by garbage collection.

Phase 6

Q 6.1

To find unreachable objects, we start from all valid references like branch heads and HEAD. From each reference, we traverse all related objects — commits, trees, subtrees, and blobs — and mark them as reachable.

A hash set can be used to store all reachable object hashes because lookup is fast.

After marking, we scan all objects in `.pes/objects/`. Any object that is not in the reachable set is considered unreachable and can be deleted.

Even if there are many branches (like 50), the traversal doesn't repeat everything each time. Since branches share history, the total traversal is roughly proportional to the number of unique commits (for example, around 100,000 commits), along with their trees and blobs.

Q 6.2

Garbage collection can cause problems if it runs at the same time as a commit.

When a commit is being created, objects like blobs and trees are written first, and only later the branch reference is updated. If garbage collection runs in between, it might not see these new objects as reachable and could delete them.

This would break the repository because the commit would point to missing objects.

To avoid this, systems like Git use locking and temporary references to make sure that objects being created are not deleted during garbage collection.